



Digital product passports

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Abstract

Digital product passports (DPPs) will become mandatory in the European Union (EU) for a wide range of product categories under the Ecodesign for Sustainable Products Regulation (ESPR). DPPs are digital artifacts that capture and share essential product data across organizational boundaries. They have the potential to enhance transparency across value chains and ecosystems, thereby supporting data-driven decision-making throughout the entire product lifecycle and unlocking new opportunities for sustainable value creation. Despite growing momentum, the design, implementation, and management of DPPs are still evolving, resulting in a vast space of design options and ambiguity. This fundamentals paper reports on a multivocal literature review to advance our understanding of this emerging phenomenon from an information systems perspective. We position the DPP against adjacent concepts, define it, develop a conceptual framework, and present a demonstrative scenario that instantiates the conceptual framework's elements. The findings have implications for DPP research and practice and lay the groundwork for achieving both regulatory compliance and enabling additional sustainable pathways.

Keywords Digital product passports · Data sharing · Digital sustainability · Circular economy

Introduction

Imagine a convenient way to get trusted information on a battery degradation profile and repair history in a format that is both readable for humans and machines. Unlike conventional cars, where mileage and age dominate value,

battery health is a key factor in determining the value of an electric vehicle. Yet, 49% of buyers remain sceptical because they cannot assess degradation over time (Venus et al., 2024). Digital product passports (DPPs) are a novel concept to overcome such information asymmetries by collecting and managing product data across organizational boundaries (Jensen et al., 2023a). DPPs provide traceable and accessible information about the product's lifecycle,

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including its origin, composition, and sustainability impacts (Walden et al., 2021), as well as create a faithful identity of uniquely identifiable tangible objects (Langley et al., 2023; van Capelleveen et al., 2023). The importance of DPPs has grown since their inclusion in the Green Deal policies and the related “Ecodesign for Sustainable Products Regulation” (European Commission, 2024). The European Commission envisions DPPs as purposeful instruments for accumulating and sourcing useful product lifecycle data from all stakeholders involved in the assembly, sale, as well as (re)use of the product or its materials, and foresees new opportunities in many markets, including textiles, electronics, construction, automotive, and chemicals (European Commission, 2024). Battery manufacturing will be the first industry legally required to implement a DPP in 2027, which is called the *battery passport* (European Parliament and the Council, 2023).

DPPs consolidate diverse responsibilities into a single artifact and constitute a promising object of inquiry for the information systems (IS) discipline, as they both draw upon and contribute to multiple research streams. *First*, the IS community can bridge the interdisciplinary nature of DPPs by reflecting on how technology and data can assist in achieving value, as known from *digital sustainability* (Kotlarsky et al., 2023) or *twin transformation* (Christmann et al., 2024). IS research has a long tradition of providing knowledge for designing and using digital applications responsibly to avoid harm and leverage their benefits (vom Brocke et al., 2013), which is important for DPPs as they have various organizational, social, and technical consequences (Schoormann et al., 2025). *Second*, inter-organizational data sharing is receiving attention in IS research (Jussen et al., 2024) to create a foundation for goals envisioned for DPPs. This demands interoperable digital architectures for managing DPP data in large collaborations. By creating new interactions among diverse actors, DPPs are expected to foster the emergence of *data ecosystems* (King et al., 2023; Möller et al., 2024). In doing so, DPPs could leverage their potential beyond regulatory compliance and craft novel value propositions (Alt, 2020; Catena-X, 2024; Heeß et al., 2024). These potentials open up opportunities to explore the role, structure, and consequences of DPPs for organizations, collaborations, and electronic markets (Petrik et al., 2025b; Ryu et al., 2024; Zeiss et al., 2021). *Third*, IS research is concerned with appropriate practices for data-driven sustainable decision-making (Krasikov & Legner, 2023). For DPP success, implementation issues, such as balancing decision rights, access, and accountability, need to be understood (Ducuing & Reich, 2023; Khatri & Brown, 2010; Machado Ribeiro et al., 2023). This requires research on *data governance* capable of handling (sensitive) data with care.

Literature has started to conceptualize DPPs (Chaudhuri et al., 2024; King et al., 2023; van Capelleveen et al., 2023).

However, except for a few studies presenting initial design and guiding principles (Heeß et al., 2024; Langley et al., 2023), prior research has predominantly focused on exploring DPPs in the context of the circular economy (Walden et al., 2021; Zhang & Seuring, 2024) or resembles around relatively isolated implementations, with missing conceptual and contextual linkages to IS (see Appendix A in the online supplement for a detailed overview of existing work). While such research is useful in understanding the purpose of DPPs, it does not clarify their fundamental nature, posing a risk of not adequately leveraging the opportunities of DPPs.

In this fundamentals paper, we address this shortcoming by adopting a multivocal literature review to (1) position the emerging class of DPPs against adjacent concepts, (2) define DPPs, and (3) structure key elements in the form of a conceptual framework. Given the relevance and growing attention of DPPs from policymakers, researchers, and industry, our work systematizes and advances the evolving body of knowledge, proposing a research agenda that can support the unleashing of their full potential in the future.

State of the art

Given the relevance of DPPs, we see emerging streams of research across disciplines (see Appendix A in the online supplement¹). Jansen et al. (2022) present an overview of existing DPP initiatives and activities from enterprises, policymakers, and research, while Zhang and Seuring (2024) and Petrik et al. (2025a) explore early application scenarios. van Capelleveen et al. (2023) describe the common features, attributes, antecedents, and consequences of DPPs. In terms of the design of a DPP, a few papers have proposed the first design principles. Heeß et al. (2024), for instance, propose design principles centered on holistic data capture, data privacy, forgery-proof data, and automated passport processing. Langley et al. (2023) propose a set of guiding principles, including a DPP requirements principle, a DPP implementation principle, a DPP impact assessment principle, and so forth. Their work is supported by technical requirements (Plociennik et al., 2022) and architecture concepts for specific domains and product types (Plociennik et al., 2024; Reich et al., 2024). Others started exploring the opportunities of integrating technologies, such as the asset administration shell (Ajdinović et al., 2024; Pourjafarian et al., 2023), or blockchain (Greiner et al., 2024; Wan & Jiang, 2025). Prior design-oriented research has mainly

¹ Table A2 in the online supplement lists the existing body of knowledge that we found in our inquiry on DPPs. Below, we will discuss some of the seminal papers and the dominant themes that shape the current discourse.

focused on conceiving data models (Berger et al., 2022, 2023; Pohlmann et al., 2024; Jensen et al., 2023a). Recent work has also been dedicated to identifying data gaps relevant to a circular economy (Çetin et al., 2023) and prioritizing data categories based on their usefulness (Jensen et al., 2023a; Pohlmann et al., 2025; Behn et al., 2025; Popowicz et al., 2025). One study explores data governance for DPPs (Ducuing & Reich, 2023), while two others describe the technical design of a data platform for governance (Ajdinović et al., 2024; Wiesner et al., 2024).

However, with some notable exceptions (e.g., Heeß et al., 2024; Langley et al., 2023), the body of literature is mostly technically oriented. Only a few researchers take a systemic view and define ecosystems on the basis of DPPs (King et al., 2023). In addition, Jensen et al. (2024) conceptualize an orchestration framework for such ecosystems, and Gieß and Möller (2025) examine the value exchanges in a DPP ecosystem. Despite these strides, previous research on DPPs is nascent and lacks a consolidated understanding of what a DPP is. This impedes the ability to anticipate its adoption (Chaudhuri et al., 2024; Schoormann et al., 2025). Our paper addresses this shortcoming by defining and conceptualizing the emerging class of DPPs, unpacking and structuring constitutive DPP elements, and demonstrating their interrelations using the example of the battery instance. We hence move beyond purely technical prescriptions of DPPs, thereby extending the conversation from “how to build a passport” to “how to conceptualize, design, and launch DPPs that realize their promised impact.”

Conceptualizing digital product passports in information systems research

A positioning of digital product passports and adjacent concepts

The goal to make physical objects traceable through *digital representations* is not new and has been studied in the IS community for years (Recker et al., 2019). As early as the 1970s, such representations were used in the construction industry to design, construct, and manage built assets (ISO, 2018). Especially with the advancement of the circular economy in supply chain management, the idea of digital records for single, identifiable, and tangible objects has gained attention (van Capelleveen et al., 2023).

A core underlying concept, typically associated with digital representations both in practice and research, is the *digital twin*. Digital twins are digital representations of real-world objects, which often rely on uni- or bidirectional data flows with the real world (Fukawa & Rindfleisch, 2023; Tao et al., 2018; van der Valk et al., 2022). The digital twin concept has developed rapidly, encompassing numerous variants (e.g., digital models, shared digital twins, or digital shadows) that

differ in the scope of how they acquire the data or the existing or non-existing reciprocal action between the physical and the digital world (Fuller et al., 2020; Haße et al., 2022; van der Valk et al., 2022). Prominent examples include Lowe’s use of NVIDIA Omniverse to create digital twins of retail stores or Tuvalu’s initiative to assess coastal impacts from physical processes (Adams et al., 2023; NVIDIA, 2025). Contrary to the DPP, the digital twin concept is agnostic regarding the specific physical entity it is supposed to mirror, which could include products, supply chains, organizations, or even human beings (Jesus et al., 2024; Strobel et al., 2022; van der Aalst et al., 2021). Accordingly, the data that goes into a digital twin is shaped by the specific domain, use case, and goals, and is often implemented in practice using the Asset Administration Shell (AAS) (Industrial Digital Twin Association, 2025). The digital twin is therefore an overarching concept, i.e., the digital representation of *anything*. The DPP, on the other hand, is a digital representation of a set of data specific to uniquely identifiable products defined by the ESPR (D’Adamo et al., 2025; Gieß & Möller, 2025). If providers of DPPs do not expand their scope beyond this, it will be highly limited in comparison to digital twins. For our paper, we follow the notion that *all DPPs are digital twins, but not all digital twins qualify as DPPs*.

Another concept involves digital representations that resemble the DPP, but which are not mandated by the ESPR. We will refer to these as *passport initiatives related to the DPP* (Çetin et al., 2023). From our perspective, the key difference is that these initiatives emerged in different industries, either through non-ESPR regulatory measures, such as the EU Regulation 2021/2117 on wine labelling (Digi-Label, 2025), or through business decisions, for instance, to ensure sustainable supply chains (e.g., Nobodyschild, 2025), before the ESPR entered into force. There is a range of similar concepts in the literature that would most likely fall under the “digital product passport” terminology today (e.g., circularity passport or recycling passport, see Zhang & Seuring, 2024). These include domain-specific Material Passports (MPs), which record data on materials, buildings, or even districts (Honic et al., 2021, 2024). By contrast, DPPs are intended as a cross-domain instrument backed by European authorities to achieve greater standardization than MPs (Çetin et al., 2023). For instance, they are to be linked to a central registry of unique product IDs, and the standardization of requirements is to be advanced by the delegated acts or pilot projects (European Commission, 2024; Cirpass2, 2025). The two concepts also differ in their reference objects: existing MP initiatives vary by representing materials, products, buildings, or building complexes, whereas DPPs are assigned to unique products, product batches, or product models (Çetin et al., 2023; European Commission, 2024). Despite these differences, DPPs and most other initiatives related to a DPP have similar objectives, and forthcoming European measures will push DPP data requirements down to the material level, as exemplified by the

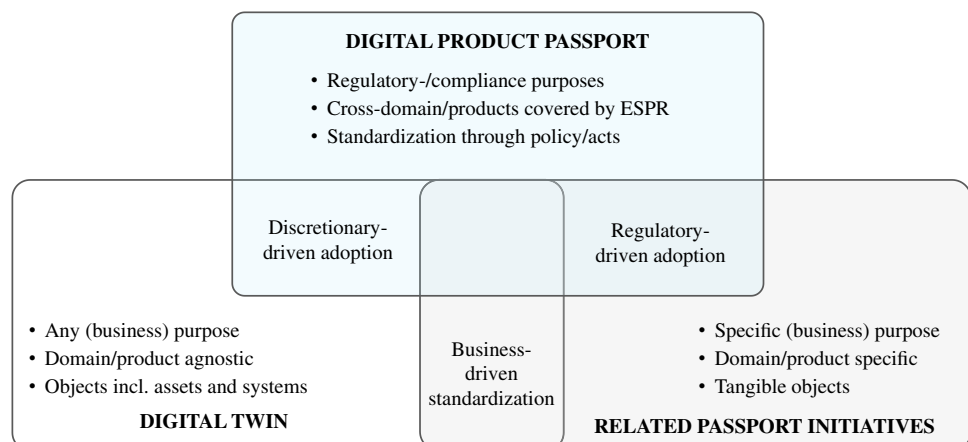
Table 1 Digital product passports and adjacent concepts

	Digital twin	Passport initiatives related to the DPP (non-ESPR)	Digital product passports
Purpose	Any purpose	Specific purpose	Identity and regulatory-driven disclosure of information
Adoption driver	Discretionary	Discretionary/regulatory	Regulatory compliance, sustainability, circularity
Standardization	Business-driven consortia	Business-driven (e.g., authenticity for wine, carbon accounting for cardboard packaging batches)	Driven by the EU regulations and delegated acts, as well as research pilots
Primary scope	Domain-/product-agnostic	Domain-/product-specific (e.g., construction, wine, luxury goods, as well as their materials)	Cross-domain, products covered by the ESPR and specialized laws (e.g., Battery Regulation)
Reference object	Asset, system, or system-of-systems	Tangible objects of varying granularity (e.g., bill of materials for a building)	Single identifiable product, product batch, or product model
Identity	Not registry-based or mandated by policy	Domain-specific identities	Unique product identified, supported by the EU registry
Data categories	Bottom-up (i.e., defined by the business needs and specific to a use case)	Bottom-up (i.e., defined by the business needs and specific to a use case)	Top-down (i.e., anchored in the ESPR, requirements defined per product group in delegated acts, refined via research pilots or domain-specific standards)
Examples	Digital twins of retail stores (NVIDIA, 2025); digital twin of Tuvalu island nation (Adams et al., 2023)	Material passports (Çetin et al., 2023), nobody's child DPP for garment (Nobodyschild, 2025), Klingele carbon accounting passports (Petrik et al., 2025a)	Battery passports (European Commission, 2023), Textile Passport (European Parliamentary Research Service, 2024)

EU Battery Regulation and the DIN SPEC 99100 (European Commission, 2023; DIN, 2025). Subsequently, we work under the presupposition that *every DPP is a digital representation of a physical object, but not every digital representation of a physical object is a DPP as mandated by the ESPR*.

Taken together, DPPs share commonalities with the concept of digital twins and passport-related initiatives, and the underlying logic behind how they work is already a reality in different domains. The DPP is primarily distinguished by its grand scale and the regulatory force behind it. This detaches existing digital representations from purely business-driven motivations and adds a distinct regulatory component. As most existing digital representations are typically not orchestrated at the EU level, the DPP, as mandated by the ESPR and specialized laws such

as the EU Battery Regulation, creates an unprecedented need for swift adoption, standardization, and, in many cases, digitization (Chaudhuri et al., 2024; Walden et al., 2021). The implementation of DPPs is not at the discretion of organizations, but is clearly mandated by the data requirements for products and the target dates for their implementation. This has severe implications: on the one hand, the ESPR forces companies to share data on an unprecedented scale, removing their ability to freely decide what, or even whether, to share that data, but on the other hand, it may fuel the advancement of data ecosystems across the EU (Chaudhuri et al., 2024; Gieß & Möller, 2025; King et al., 2023). Table 1 shows a comparison of the three different digital representation concepts, and Fig. 1 visualizes them by emphasizing their overlaps and distinctive aspects.

Fig. 1 Comparison of different concepts

A definition of digital product passports

The DPP is an emerging concept whose definitive data category, governance, and collaboration requirements (e.g., Jensen et al., 2023a; Ducuing & Reich, 2023) remain unsettled. Policy and scholarship converge on the idea that a DPP aggregates data from the value chain into lifecycle information and makes it accessible to value chain actors and consumers, which leads to the creation of a data-driven value ecosystem (European Commission, 2020; King et al., 2023; Gieß & Möller, 2025). The starting point to define this concept is the ESPR, which proposes that a DPP “*means a set of data specific to a product that includes the information specified in the applicable delegated act adopted pursuant to Article 4 and that is accessible via electronic means through a data carrier*” (European Commission, 2024).

Given the wide range of potential applications for DPPs, the ESPR definition is rather broad and open to interpretation. This has prompted researchers and practitioners to extend, build upon, or adapt their definitions (see Appendix A and B in the online supplement). Fully aware that the discourse on DPPs is still evolving, in this fundamentals paper, we propose a definition that is informed by existing definitions and the ESPR, and reflects the existing advancements in policy, academia, and practice (see Appendix B in the online supplement). Subsequently, we propose the following definition:

A Digital Product Passport (DPP) is a policy-driven digital representation of a unique, identifiable product, product batch, or product model that is mandated by the ESPR to capture, organize, and disclose mandated product data across organizational and ecosystem boundaries throughout the product lifecycle via a data carrier and interfaces, acting as a digital artifact to support regulatory compliance.

Following this definition, the EU intends to use DPPs to foster the collection of product-related data throughout the product lifecycle to make it shareable and usable by an ecosystem of actors via physical data carriers and digital interfaces to meet recurring information needs for sustainable action (Voulgaridis et al., 2024; European Commission, 2024; Gieß & Möller, 2025). The nature of DPPs aligns with the properties of digital artifacts, which means that DPPs are open, reprogrammable, interactive, and distributed (Kallinikos et al., 2013). These properties enable economic operators who place tangible objects captured by the ESPR on the market to ensure that the data is “accurate, complete, and up-to-date” (European Commission, 2020), and, together with other value chain actors, to co-create a single point of truth—a faithful representation of a uniquely identifiable tangible object to inform other actors (Recker et al., 2019). Although DPPs hold promise to increase data

sharing (Chaudhuri et al., 2024; Gieß & Möller, 2025; King et al., 2023), leveraging this potential requires a deeper understanding of their constitutive elements. Therefore, we extend the definition with a conceptual framework for DPPs.

A conceptual framework for digital product passports

DPPs are more than just barcodes or electronic labels, as they can contain a significantly higher density and complexity of data from the lifecycle of a product or thing (Daphne, 2024). They represent a higher-order concept independent of a specific instantiation and can contain data on any product or tangible object (King et al., 2023). A variety of interoperable instances coexist to cope with the complexity of real-world tangible objects. Their data can be made available to both actors and other systems in a machine-readable format. Instances of such DPPs are the *battery passport* or the *textile passport*, which incorporate lifecycle data unique to individual batteries or garments (Berger et al., 2022; Götz, 2022; King et al., 2023). Figure 2 shows the advancing instantiations of a DPP, highlighting differences in scheduling and data models. This demonstrates the diversity of DPPs and the captured data and illustrates how DPPs will be tailored to a particular domain context.

Given the multitude of elements involved in the realization of DPPs and the resulting equifinality of implementing DPPs, we complement the definition with a conceptual framework, which draws on the ESPR and the literature review sample.² The framework (see Fig. 3) provides a structured view of the constitutive elements, shaded grey, and their non-exhaustive instantiations in white. It specifies the relationships between these elements and remains flexible to accommodate new elements that may emerge from future regulations. In the following, we detail each element of the conceptual framework, which together constitute the building blocks of future DPPs.³

Actors

A DPP is used by different actors, i.e., stakeholders at the organizational level involved in a product’s lifecycle, who collaborate based on data for mutual benefits (Berger et al., 2022; Jensen et al., 2024). Typical actors include consumers who purchase tangible objects and different value-adding partners in the supply chain, such as manufacturers, suppliers, and recyclers. These actors are expected to source

² Due to space restrictions, we use example references in the result section; see the full sample in Appendix A in the online supplement.

³ The supplementary file contains an expanded list and a glossary of all elements of a DPP.

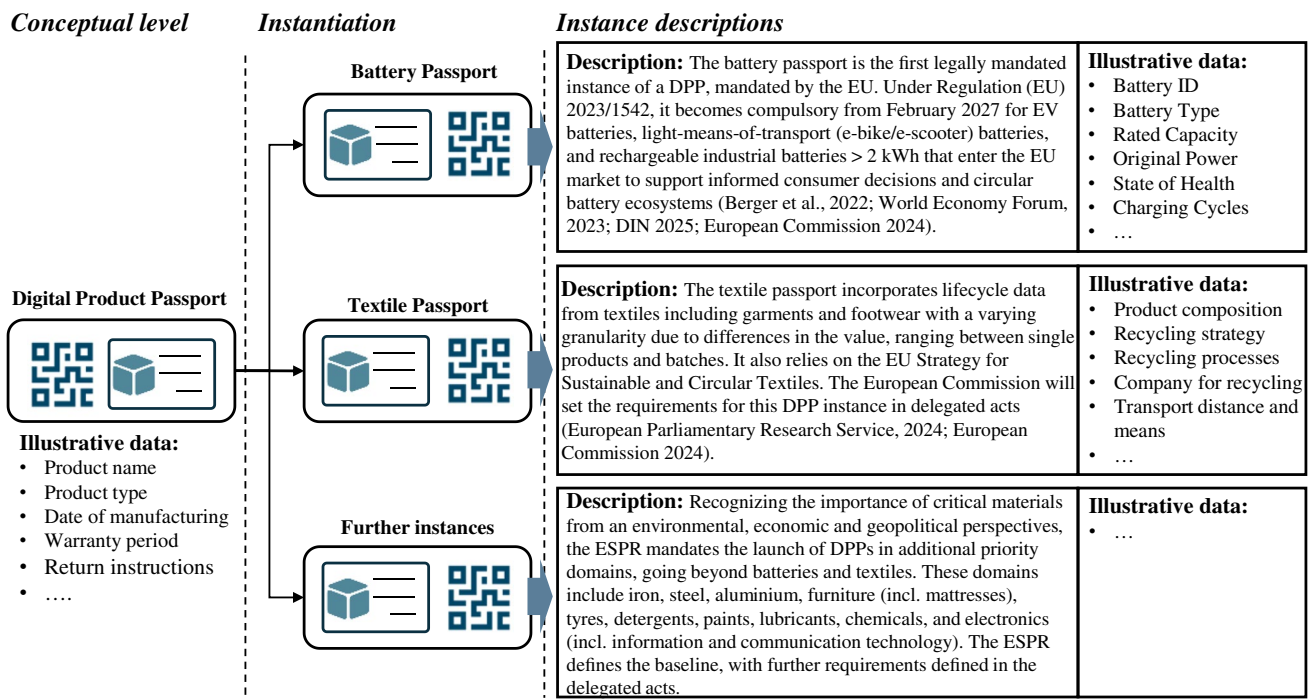


Fig. 2 Instantiations of digital product passports

static upstream and dynamic downstream data (Jensen et al., 2023a). Static upstream data, such as data concerning the bill of materials, is often used to support decision-making, neglecting the importance of the dynamic use-phase. This

use-phase data can inform relevant actors about the product's journey to optimize decisions, such as those related to its end-of-life treatment (Jensen et al., 2023a). Other partners are specialized service providers (e.g., repair,

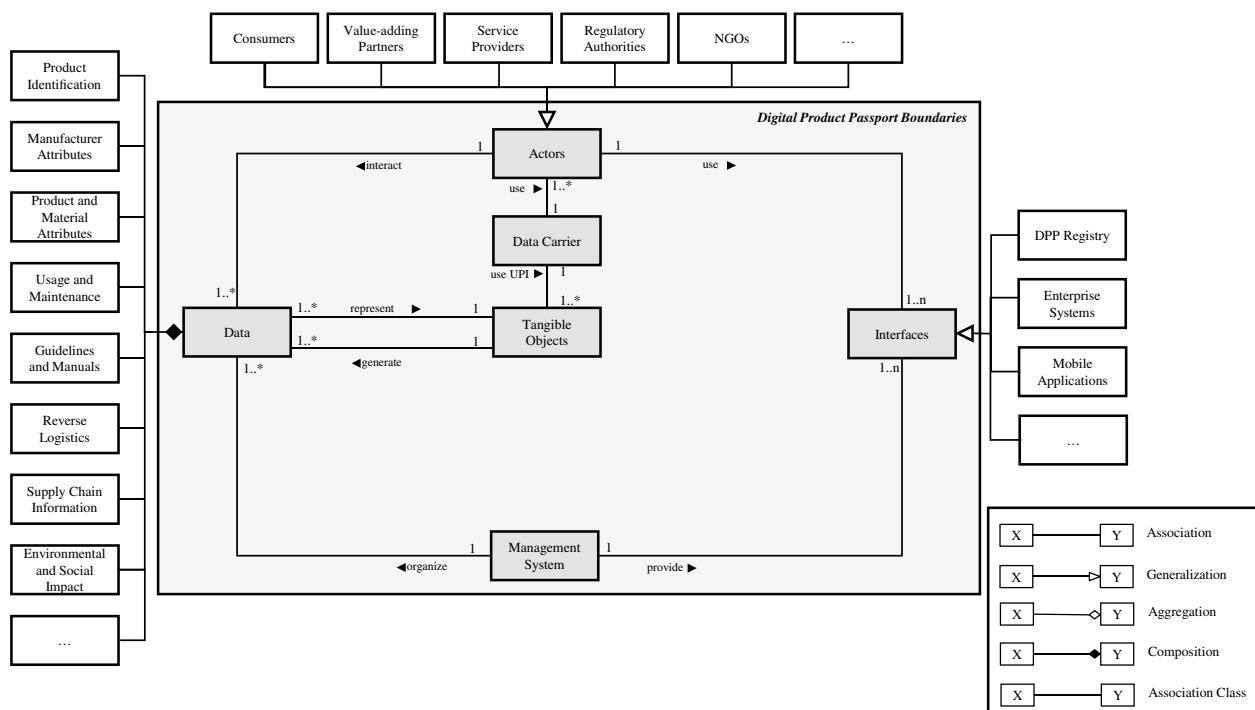


Fig. 3 Conceptual framework of digital product passports

remanufacturing, logistics, research), regulatory authorities, and NGOs. They interact with the DPP to benefit from the information available (Jensen et al., 2024; King et al., 2023). Digital service providers offer DPPs as a service and help to store and share data (King et al., 2023; Luscuere & Muhlall, 2018).

All actors generally operate within a framework of laws, regulations, standards, and guidelines that determine the creation, implementation, and management of DPPs. It ensures that DPPs are used in a way that meets legal obligations and promotes transparency, accountability, and sustainability (Plociennik et al., 2022; European Commission, 2024; King et al., 2023). Hence, DPPs are also expected to be used for applications that go beyond mere compliance, bearing the potential of creating environmental, social, or economic value for the actors involved (Berger et al., 2022; Jansen et al., 2022; Pohlmann et al., 2024; Reich et al., 2023).

Tangible objects

A DPP is always associated with a tangible object. It is the entity for which the DPP collects and incorporates lifecycle data. Under the ESPR, a tangible object can have different granularity levels, such as batches of low-value items (e.g., textiles, packaging), individual products, or product models. Considering modular products and capital goods, DPPs can be created for individual modules to support remanufacturing at the module level (Götz, 2022; Stretton, 2022).

Interfaces

Since products are rarely manufactured by isolated companies but instead integrated into value chains or ecosystems, it is essential to collect data from all upstream and downstream actors who have touchpoints with the product to feed it into a DPP throughout the product lifecycle to enable a holistic data model (Heeß et al., 2024; Piétron et al., 2023). To achieve this, *interoperability* is key. Information access is not only limited to value chain actors but also serves secondary actors from policy (e.g., national authorities), economy (e.g., business associations), or social NGOs (King et al., 2023). Therefore, a DPP requires the ability to interact with various technical systems. Information is accessed via interfaces. For instance, data can be exchanged between DPPs and *enterprise systems* of the value chain actors (Langley et al., 2023; Monteiro et al., 2024; Petrik et al., 2025a; Plociennik et al., 2024). In addition, certification services can be used to pull data to ensure compliance with relevant standards or certify data via interfaces (Langley et al., 2023). In the future, the EU Commission intends to launch and operate a superordinate *DPP registry* at the European level to verify each DPP's authenticity (European Commission, 2024). At the individual level, consumers interact with the DPP

primarily through *mobile or web applications*. This means that DPPs and their instances have both a human-machine and a machine-to-machine interface. Through these interfaces, a robust exchange of data can be facilitated to inform stakeholders, promoting transparency, traceability, and faithfulness throughout the product lifecycle (Walden et al., 2021; van Capelleveen et al., 2023; European Parliamentary Research Service, 2024). Although formal interface specifications are not defined by the EU authorities yet, delegated acts and the outcomes of pilot projects will likely define them, drawing on modern web service standards, such as Representational State Transfer (RESTful) API design and JavaScript Object Notation (JSON) schema (Pourjafarian et al., 2023; Ajdinović et al., 2024; Cirpass, 2024; Cirpass2, 2025).

Data carrier

The data carrier is also used by the actors and connects the 1..n of tangible objects over the globally valid *unique product identifier* (UPI) to the DPP. The data carrier must be physically identifiable on the product, in its package, or any supporting paperwork, and comply with ISO/IEC 15459:2015. To provide low-barrier access to a DPP, various track-and-trace technologies are tested, including Radio-Frequency Identification (RFID), Quick Response (QR) codes, barcodes, and Near-Field Communication (NFC) (Giovanardi et al., 2023; Roio et al., 2024; Gligoric et al., 2019). While these methods differ in durability, resistance to physical entropy, and ease of scanning, they all establish a secure link between tangible objects and their digital representation. In addition, the data carrier should resolve to a uniform resource locator (URL) on the internet and be connected with the existing identifiers, such as the European Article Number (EAN) or other GS1 standards (Plociennik et al., 2022).

Data

The data in DPPs is granular and captures information about individual products (Langley et al., 2023). It is dynamic, being updated throughout the lifecycle as new information becomes available. The tangible object can trigger and generate these updates, resulting in a structured aggregation of information that is useful to the aforementioned actors (Jensen et al., 2023a; Popowicz et al., 2025). In particular, each DPP comprises multiple data entries, where each entry represents an individual piece of information, such as date of manufacture, materials used, maintenance records, or environmental impact metrics (Behn et al., 2025; Berger et al., 2022; Jansen et al., 2022). Because regulations or actors' information requirements evolve over time, DPPs need to have an extensible data model, conceptually building

a knowledge graph (Jensen et al., 2023a; Cirpass, 2024). To foster inter-organizational data sharing, the DPP data needs to be machine-readable, structured, searchable, and built using an interoperable format based on open standards (Berger et al., 2022; Plociennik et al., 2022; Jensen et al., 2023b; Piétron et al., 2023). Although data standards have not yet been set for many product domains, the ESPR may draw on existing standards, such as the AAS, in the delegation acts. The AAS incorporates the possibility to create sub-model templates for any tangible object category to facilitate consistent data models and semantics across organizational and ecosystem boundaries (Ajdinović et al., 2024; Pourjafarian et al., 2023).

Management system

DPPs require an overarching management system that serves as a data platform to orchestrate the collection, storage, and processing of 1..n data from multiple sources (Cirpass, 2024; Wiesner et al., 2024). This system automates the creation of data models for each DPP, updates data attributes, and adapts the information model to meet evolving stakeholder or regulatory needs (Langley et al., 2023; Voulgaridis et al., 2024). To reflect changes in product ownership or multiple lifecycles, the management system also needs to process data updates and orchestrate transitions of a DPP to new owners (Heeß et al., 2024; Reich et al., 2023). Because these updates may demand specific access rights, the system needs to support flexible data storage and retrieval for authorized actors. NoSQL databases, for instance, are well-suited for flexible storage in contrast to relational databases or file structures. In terms of data access, consumers or regulatory authorities typically need read-only permissions (e.g., for reparability or detecting counterfeits), while write permissions are reserved for tasks such as carbon accounting or updates of product specifications. Accordingly, it provides interfaces for the actors. A management system also needs to offer data governance and security capabilities to enable encryption and access control, resolving the issue of who is authorized to do what (Piétron et al., 2023; Wiesner et al., 2024). To enable trustworthy data governance, decentralized approaches are recommended, allowing for adaptive and transparent administration (Ducuing & Reich, 2023; Greiner et al., 2024; Heeß et al., 2024). The management system must also be able to deactivate an existing DPP or issue a new one, eventually at the component level, when the original object is decommissioned but its parts are reused (Potting et al., 2017). To ensure trust in the integrity of the data in a DPP from the user's perspective and to protect data against manipulation in a competitive setting of free markets, these capabilities are critical for the value chain actors to share their data and co-create faithful digital representations in a DPP (Heeß et al., 2024; Wan & Jiang, 2025).

These management systems will likely be domain-specific (R-Cycle, 2024; Siemens, 2024b).

Demonstrating the conceptual framework for battery passports

The battery domain provides a purposeful context for demonstrating how each framework element can be instantiated because battery passports are the first legally mandated DPP instance in the EU. The automotive industry is transitioning to electromobility, using batteries for which many data points are available. However, data gaps and information asymmetries prevail between the globally dispersed value chains and consumers in the secondary battery markets (Serna-Guerrero et al., 2022; Venus et al., 2024). By explicating how the instantiated elements apply in this domain, we demonstrate how each element can be adopted and highlight the mechanisms through which DPPs ensure regulatory compliance and generate additional value.

When batteries, as *tangible objects*, reach their end-of-life stage, choosing the right end-of-life approach involves complex decisions (Berger et al., 2022; Shahjalal et al., 2022) by car manufacturers, battery manufacturers, recyclers, or resellers owning electronic marketplaces and other actors involved in battery take-back systems (Gieß & Möller, 2025; Serna-Guerrero et al., 2022; Wellten et al., 2025).

These *actors* require rich and faithful *data* about a product's composition, residual quality (e.g., capacity, charging rate, negative events), disassembling options, and many more (DIN, 2025; Hoppe et al., 2024). Without lifecycle data, laborious activities, such as disassembling and visual inspection, are performed manually, which makes handling of end-of-life batteries uneconomical and prone to errors (Andersen et al., 2023). A *battery passport* (i.e., a *DPP*) creates a persistent identity that unambiguously links each battery to its historical data, improving the efficiency of disassembly and testing (Berger et al., 2022). Aggregating data on cell chemistry, prior usage, and maintenance helps make decisions on end-of-life options, recycling processes, and secondary materials (Harper et al., 2019; Reich et al., 2023; Yu et al., 2022).

Interfaces enable economic operators and battery suppliers (i.e., upstream partners) to source data on battery chemistry and critical raw materials (DIN, 2025) from their enterprise systems into the DPP to satisfy the information needs of recyclers and dismantlers (i.e., downstream partners) regarding battery composition, enabling them to disassemble batteries without destroying them. This would enable downstream partners to perform more sustainable circularity principles than recycling (Gieß & Möller, 2025; Monteiro et al., 2024; Plociennik et al., 2024; Rettenmeier et al., 2025). Bidirectional interfaces can enable a reverse flow of

aggregated data from the DPP into the enterprise systems for processing and reporting product carbon footprints. As accounting for upstream and downstream emissions (i.e., Scope 3) is expected to become mandatory, it creates information needs on carbon emissions from each process step of each partner in the value chain (Emborg et al., 2023; Götz, 2022). DPPs serve as a distributed and open source of primary data, reducing reliance on averaged secondary data (Ströher et al., 2025) and grounding a faithful accounting system (Heeß et al., 2024; King et al., 2023). The regulatory authorities and NGOs can also use these interfaces to facilitate remote queries for compliance purposes (Gieß & Möller, 2025; King et al., 2023).

Since consumers cannot easily assess battery health or locate trustworthy take-back options, binding a persistent battery identity to verified state-of-health data is crucial to mitigate consumer information asymmetries in secondary markets and enable battery reuse. Consumers can assess this data via physical *data carriers*, which also provide information on certified recycling partners or material attributes (Piétron et al., 2023; Zhang & Seuring, 2024). Such transparency can help value chains to mobilize consumers to return batteries to their original resource loops (Donner et al., 2021; Recker et al., 2024), supporting circularity (Berger et al., 2022; Pohlmann et al., 2025).

To enable value chain actors to update the data of a DPP and grant sufficient data access to customers and authorities, a *management system* is required. Such systems must manage access to sensitive data, which a DPP may contain (DIN, 2025), and ensure data integrity (Ducuing & Reich, 2023; Heeß et al., 2024). In addition, if components, such as battery packs, are reused in other products after disassembly, this should be reflected in the DPP of the newly assembled product (Heeß et al., 2024; Wiesner et al., 2024; Gieß & Möller, 2025; Langley et al., 2023; Jensen et al., 2023a). Management systems must also be able to add new data categories to the DPPs (Jensen et al., 2023a; Cirpass, 2024; Wiesner et al., 2024), enabling DPPs to adapt to evolving regulations, for instance, by including additional data on ethical labor practices in support of responsible sourcing (Berger et al., 2022; King et al., 2023; Uusitalo et al., 2023).

Implications for research and practice

The discourse on DPPs and their implications has been primarily driven by political authorities and practitioners (Berger et al., 2022; European Commission, 2024; Global Battery Alliance, 2020; World Economy Forum, 2023), while the scholarly knowledge base has remained fragmented (see heterogeneous definitions in Appendix

B in the online supplement). In this fundamentals article, we advance this emerging artifact class by (1) positioning DPPs against related digital representation concepts, (2) proposing a definition that complements the ESPR and adds an IS perspective, and (3) crafting a conceptual framework with six constitutive DPP elements and their interrelationships. The framework aims to establish a shared understanding by connecting theory, regulations, and practice. It enables various stakeholders to talk about the same thing and distinguish regulatory-driven DPPs from other digital representations. By demonstrating the framework with the case of the battery industry, we illustrate how the key DPP elements are used in real-world settings to leverage their potential.

Implications for digital sustainability

Consistent with the idea of digital sustainability (Kotlarsky et al., 2023), DPPs present *digital artifacts that draw on data to contribute to sustainable development*. Fostering a mutual understanding of DPPs, our results support economic actors in avoiding societal sanctions and maintaining stakeholder legitimacy by operating in a legally compliant manner (Carroll, 2016). A comprehensive collection of aggregated data accessible across domains and organizations enables further advancements toward economic, ecological, and social goals. Therefore, it is important to understand the opportunities offered by DPPs (Hoppe et al., 2024; Ryan & O'Malley, 2016).

DPPs serve as exemplary digital artifacts that reflect evolving research at the intersection of *regulations and technology* (e.g., Seidel et al., 2024; Kurtz et al., 2025). Our conceptual framework captures key elements and relationships, informing design decisions aligned with sustainability regulations. This not only represents a contextual shift for IS in practice (Pfeiffer et al., 2024), but also serves as a fundamental baseline for economic actors in the domains affected by the ESPR to remain on the market.

Besides the positive impact, potential *tensions and unintended consequences* of DPP adoption need to be considered to prevent backfiring and undermining sustainability goals. For example, as pointed out in the Demonstration section, policymakers consider DPPs as artifacts to increase product transparency, which can reduce transaction costs by decreasing information asymmetries, but bear additional risks, such as high implementation costs and the risks of data breaches (Schoormann et al., 2025). Our work helps organizations interacting with DPPs and designers to better reflect on such issues by matching them to the DPP key elements, as well as identifying causes and ways to address them.

Implications for interorganizational data sharing and data ecosystems

When reflecting on the increasing significance of data and data sharing to improve today's globally dispersed value chains, it is imperative to include DPPs in these discussions (Alt, 2020; Möller et al., 2024). DPPs can serve as a *boundary object to integrate data* across actors and establish new collaborations and ecosystems (Gieß & Möller, 2025). It is well known that, especially in business-to-business domains, data is often seen as critical for one's competitive advantage and is not shared (Jussen et al., 2024). Inter-organizational data sharing is affected by trust issues and has already led to significant drawbacks for other digital artifacts. For example, digital industrial platforms continue to struggle to convince actors to share data critical to competition with other ecosystem actors via centralized platforms (Möller et al., 2024; Springer et al., 2025). Decentralized approaches to data exchange, such as blockchain or dataspace connectors (Centobelli et al., 2022; Gieß et al., 2024), are still evolving to become relevant standards. DPPs can serve as vessels for data sharing and aligning stakeholders around common goals (Jensen et al., 2024).

We argue that DPPs should be seen as a new *type of data infrastructure* (e.g., Parmiggiani et al., 2024), for which data curators in different organizations need to collect, prepare, and make use of data. Unlike earlier digital representation initiatives, the regulatory mandate to disclose data reduces traditional barriers to data sharing but also imposes obligations on firms to provide standardized, high-quality, and auditable data. While organizations may initially perceive this as a compliance burden, the obligation to source DPPs also creates opportunities to leverage DPPs to create value that goes beyond regulation. We expect that this will have implications for data sharing and data ecosystems between organizations. By fostering new forms of inter-organizational collaboration, DPPs will likely require new data ecosystems with new types of actor interactions. This creates a potential starting point for generativity driven by (data) network effects in supply chains that were previously insular (Wu et al., 2024). However, as any digital infrastructure is deeply embedded in existing structures that actors may be reluctant to open (Henfridsson & Bygstad, 2013), DPPs can be affected by threats from individual actors who undermine collective benefits from data sharing (Mindel et al., 2018).

Implications for data governance

Adopting DPPs may enable organizations to harmonize their *data practices* to jointly manage the digital representations of uniquely identifiable products. Data sharing is contingent upon effective governance measures that protect sensitive data but still enable compliance and collaboration. For

instance, it is crucial to consider interoperability issues and validation methods for DPP adoption (Ducuing & Reich, 2023; Vial, 2023).

Consistent with existing research (Ducuing & Reich, 2023), our framework implies that DPPs raise fundamental questions of decision rights and data sovereignty. Decision rights need to be clearly allocated to the different data categories and actor roles. At the same time, regulatory disclosure requirements simultaneously undermine firms' claims to data sovereignty by reducing their control over who can access and use data in the exposed categories. This will also restructure informational power along the value chain. Since governance should align with the system architecture, our conceptualization of management systems helps to understand which capabilities are needed for data management, reflecting known data governance dimensions (e.g., clarification of data as an asset, quality, interoperability, access, and lifecycle of data) (Khatry & Brown, 2010). Although regulatory disclosure will increase the availability of product data, the actual use also demands new data reconciliation practices (Krasikov & Legner, 2023). In view of these issues, one can expect that data governance will be reconceptualized by regulatory authorities and other relevant actors (Seidel et al., 2024).

Avenues for future research on digital product passports

Value networks and ecosystems, data sharing, and sustainability have been core topics in the field of electronic markets (Alt, 2020; Möller et al., 2024; Ryu et al., 2024). DPPs are an emerging class of digital artifacts that address those topics and provide promising avenues for the electronic market community. By reflecting on the aforementioned implications, we propose guiding questions for future research, including suitable methods and theories, to address the challenges and questions raised about DPPs (see Table 2). Those are structured along the three perspectives for DPP data inputs, DPP design and governance, and DPP impact. We are confident that our work helps to explore and leverage the benefits of DPPs for legal compliance and beyond.

First, to craft and provide a DPP, the *data input* needs to be considered (Gieß & Möller, 2025). With the growing dynamics and complexity of today's business ecosystems, networks, and value chains, it has become even more challenging to disclose the traceability of interwoven product pathways. Thus, it is crucial to have faithful data sources and practices to ensure the validity of such passports and contribute to a certain degree of trust, enabling different actors to provide their data (Heeß et al., 2024). The question of how this data is stored needs to be clarified for each domain to comply with the principles of data sovereignty, which may

Table 2 Future research avenues regarding the challenges of digital product passports

Perspectives	Challenges and possible research questions	Illustrative methods and theories
On the (data) input and creation of DPPs	• How can data sources be checked and authenticated to ensure that data for DPPs is not compromised or manipulated? • What mechanisms help DPPs transparently disclose the origin of the data? • How can DPPs ensure data sovereignty and mitigate trust issues in inter-organizational (data-sharing) collaboration beyond Europe?	• Boundary object theory (DPPs as boundary objects to establish a shared understanding of data) • Representation theory (to faithfully represent real-world phenomena) • Signaling theory (to build trust between several DPP actors sharing sensitive information)
On the design and governance of DPPs	• How to design a situational DPP instantiation? • How to motivate participation by DPP design? • How can architectures for managing DPPs be designed? • What technologies can be used to design trustworthy and sustainable DPPs?	• Cognitive fit theory (to create DPPs that match the user's cognitive processes) • Design science (to create digital passports with minimal negative impact, green DPPs) • Reference modeling (to build configurational semantic foundations of DPPs)
On the use and impact of DPPs	• How can DPPs advance an organization's transition toward more sustainability, such as through the circular economy and sustainable product design? • How can DPPs facilitate consumers' sustainable awareness and behavior? • What are the ethical implications of DPPs? • What impact can DPPs have on compliance, resiliency, or fair value distribution? • How can DPPs create new value streams, such as via synergies between DPPs and platforms? • How can DPPs support sustainability reporting?	• Adoption theory (to explore how users and organizations accept and employ DPPs) • Ethics (to consider positive and unintended consequences of implementing and using DPPs) • Stakeholder theory (to incorporate interests from different actors involved in and affected by DPPs)

be at odds with the concept of DPPs (Ducuing & Reich, 2023; Scherenberg et al., 2024). For research, this offers the potential to explore how trust problems can be mitigated and how actors with differing interests can be aligned to source a DPP, considering the specifics of each domain and the boundaries imposed by the product type (Jensen et al., 2024; Piétron et al., 2023).

Second, questions can be raised about the *design and governance of DPPs*. This includes aspects such as concrete implementation examples (e.g., aspects of the visual appearance), general architectures and blueprints, and application scenarios to adopt DPPs in real-world use cases. Consider, for instance, the mechanisms for managing DPPs whose tangible objects reach the end of life. This will require management systems that can issue DPPs for certain sub-components after remanufacturing or transfer ownership to the new owner of the tangible object during reuse (Wiesner et al., 2024). As purposeful designs of such systems are still in their infancy, it is an exciting opportunity for research to explore and support their design. Given the competitive relevance of the data stored in DPPs, it will be necessary to determine how this data can be protected while being used across companies (Ducuing & Reich, 2023; Scherenberg et al., 2024). These questions can be pursued with, among others, design science research that aims to provide

purposeful design solutions, allowing for the creation and evaluation of passports.

Third, exploring the *DPP's impact* on economic, environmental, and social issues is a worthwhile avenue. DPPs may pave the way for future research on advancing assessment techniques that capture social issues (e.g., social lifecycle assessment), as well as complement existing methodologies that shed light on complex social interactions and social impacts of products across their entire lifecycle (Hahn, 2022). It is also interesting to explore how DPPs influence the structures and dynamics of platform ecosystems. For instance, examining the routines by which DPPs facilitate collaboration and innovation among platform ecosystem actors can deliver important insights into the symbiotic use of data by platforms and DPPs (Ströher et al., 2025; Wulfert et al., 2024). Since platforms increasingly incorporate artificial intelligence and advanced analytics capabilities, they help platform owners, value chain actors, or regulators accumulate knowledge and extract insights from data processing from DPPs (Alt, 2021). Therefore, we encourage future research to explore how DPPs can act as complementarities to platforms or if other types of relationships could emerge between platforms and DPPs that would positively impact sustainability (Alt, 2020; Petrik et al., 2025b). This would enable the ontological reversal concept, in which digital

artifacts are created first to subsequently shape the real world (Baskerville et al., 2020; Kotlarsky et al., 2023).

DPPs are likely to enable the exploration of new approaches for benchmarking sustainability performance concerning standards or laws and, based on the calculation, serve as a starting point for improving business routines and organizational structures. DPPs may assist the fair distribution of financial gains within ecosystems or value chains (e.g., resistance against monopolies and dominant players through transparency). Lastly, DPPs constitute a central policy instrument in the EU's transition from linear to circular product lifecycles. They are part of the EU's Ecodesign for Sustainable Products Regulation (ESPR) to boost the Circular Economy Action Plan (European Commission, 2024), given that the implementation of circularity principles continues to lag behind expectations (Corvellec et al., 2022). With increasing transparency, a DPP is likely to promote a more systemic perspective among stakeholders, leading them to collaborate on improved data to enhance overall value chain benefits (Jäger-Roschko & Petersen, 2022; Piétron et al., 2023). Thus, it would be beneficial for IS scholars to explore possible interventions caused by using DPPs, such as the consequences for circularity.

Conclusion

DPPs are digital artifacts that will be in constant flux, capturing different data about tangibles over their lifecycle—almost like a Curriculum Vitae. As datafication continues, we expect that DPPs will profoundly change inter-organizational data sharing and enhance digital traceability by creating a single point of truth at the product level. DPPs can be configured to protect sensitive data by differentiating between unauthorized access and offering availability in an anonymized form. Implementing appropriate data governance with stakeholder consensus can motivate organizations to share data they might otherwise keep to themselves, thereby supporting the widespread adoption of DPPs. Shared data serves as a foundation for comprehensive sustainability reporting of complex product pathways and enables more informed decision-making, including the selection of products that are aligned with certain values (e.g., customer perspective) or the transformation toward circular value chains and ecosystems (e.g., organizational, network perspective). This fundamentals paper advances this new class of digital artifacts by synthesizing existing knowledge to provide a comprehensive definition and a conceptual framework that creates a shared understanding of a DPP's key elements. With the future research avenues suggested in this paper, we hope to spark fruitful discourse and opportunities for DPP adoption in both research and practice.

This paper emphasizes the vast potential of DPPs and their mechanisms for creating digital representations of unique, identifiable tangible objects, and for sharing data to support sustainable action. We believe that by responding to the growing demands for data distributed across organizations, DPPs are here to stay.

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Declarations

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